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The Ethics of Automation

What should humans delegate—and what must remain answerable?

Sirui Han (韩斯睿) · Assistant Professor, HKUST



Physical control · AF447



Engineering operations · MTR



Delegated coding · Copilot

Images: P. Kierzkowski, F-GZCP (CC BY-SA 3.0) · MTR paper, Fig.2 (2014) · GitHub product visual (2025)

Three deployments, three kinds of authority

The same ethical question travels across physical control, public administration and conversation.



AF447 · 2009

Unreliable airspeed led to autopilot disengagement. The crew did not recover from the stall.



Robodebt · Australia

Annual income averaging was used to infer fortnightly welfare overpayments.



Duplex · 2018

A narrow-domain assistant called businesses to arrange appointments for a user.

Who could intervene —and who remained answerable?

Photos: F-GZCP, Kierzkowski / CC BY-SA 3.0 · Commission report volumes, Guardian · Duplex 2018 demo, YouTube repost.

Evidence: BEA (2012) · Robodebt Commission (2023) · Google Research (2018)

Analyse the evidence behind automation

Graduate-level analysis connects ethical reasons to the systems and institutions that actually operated.

Reconstruct the real workflow

Identify the automated function, the authority boundary, and the people able to intervene.

Interpret measured outcomes

Check risk, workload and productivity metrics against their study design and baseline.

Evaluate engineering controls

Use incident reports and implementation evidence to distinguish working controls from absent ones.

Defend the ethical conclusion

Explain accountability, recourse and benefit distribution without treating efficiency as sufficient.

Evidence test: which document supports the claim, and what does it not establish?

Syllabus alignment: CILOs 1 –5 · Evidence-based case analysis, application and reasoned oral argument.

Knight Capital: delegated authority to trade

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DOCUMENTED 2012 MARKET-ROUTING FAILURE · SEC SETTLED ORDER



Reuters / NYT: Knight traders at NYSE, 2 Aug 2012 (context photo).

MARKET-FACING AUTHORITY

SMARS automated equity router
Production code generated and submitted child orders into the market.

Incoming parent order → enters route

Child orders
Could execute before diagnosis.

212 parent orders → **4 million** executions in **154** stocks

More than 397 million shares in about 45 minutes.

Delegation became unsafe when market-facing authority outpaced the documented ability to monitor and stop SMARS output.

Sources: SEC Order (2013), p. 6 · Reuters photo via NYT · Findings settled without admission or denial.

MTR ETMS: automation by function

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2014 IMPLEMENTATION ACCOUNT · FUNCTIONAL MAPPING, NOT A CLAIM ABOUT CURRENT CONFIGURATION



Historical ETMS interface, 2005 paper Fig. 2 — a visual record, not a verified 2014/2026 screen.

Four functions; one approval boundary

2014 account: proposal, checking, generation and execution. Validation is not approval.

1 · Input Staff and contractors propose work in web forms.

2 · Check Rules flag conflicts; submitters can revise.

3 · Generate AI produces a weekly plan three weeks ahead.

4 · Authorise Weekly planners vet difficult contentions and give final approval.

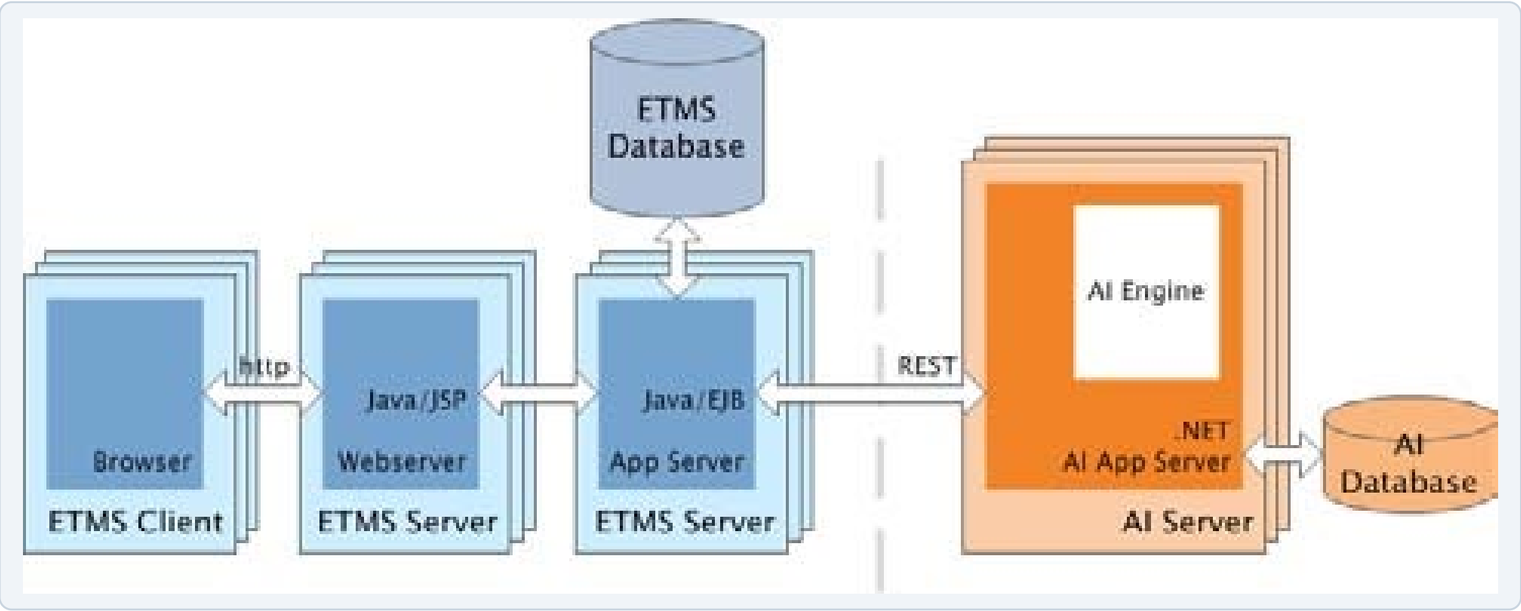
Feasible schedule ≠ authorised schedule.

Sources: IAAI/AAAI 2014 ETMS account (workflow) · CityU IAAI 2005 Fig. 2 (historical interface).

MTR ETMS: capability is not authority

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DOCUMENTED PLANNER / SYSTEM BOUNDARY · THE DIAGRAM DESCRIBES A 2014 IMPLEMENTATION



Published ETMS application architecture, 2014 paper Fig. 3 — technical separation is not a delegation-of-authority map.

What the published record supports

System capability
Rule checks, resource/conflict checks, weekly schedule generation.

Human approval boundary
Planners vet, negotiate difficult conflicts, and give final approval.

Not established by the paper

No named approver or re-approval rule for every reschedule.

Validate · rule-compliant request is not an approval.

Generate · feasible plan is not an authorised plan.

Vet · human meeting is the documented final boundary.

Source: MTR ETMS implementation account (IAAI/AAAI, 2014) · Post-reschedule re-approval is not reported.

Knight: control boundary broke at three phases

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ACTUAL SEQUENCE · STAGED RELEASE FROM 27 JULY 2012

DEPLOYMENT STATE: EIGHT SMARS SERVERS

Release intent: Power Peg → RLP on all eight production servers.

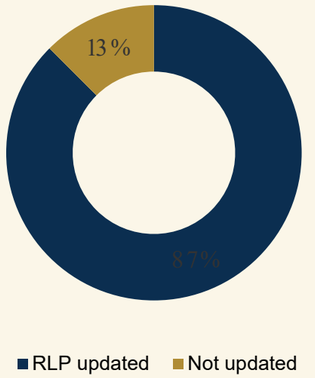
SMARS 1 · RLP	SMARS 2 · RLP	SMARS 3 · RLP	SMARS 4 · RLP
SMARS 5 · RLP	SMARS 6 · RLP	SMARS 7 · RLP	SMARS 8 · legacy

↓

At market open, a repurposed flag reached the non-updated server and activated defective Power Peg.

Boundary lesson: verify the production state before activation; then monitor final market-facing output.

DEPLOYMENT COMPARISON



7 / 8 received the new RLP code and processed eligible orders correctly.

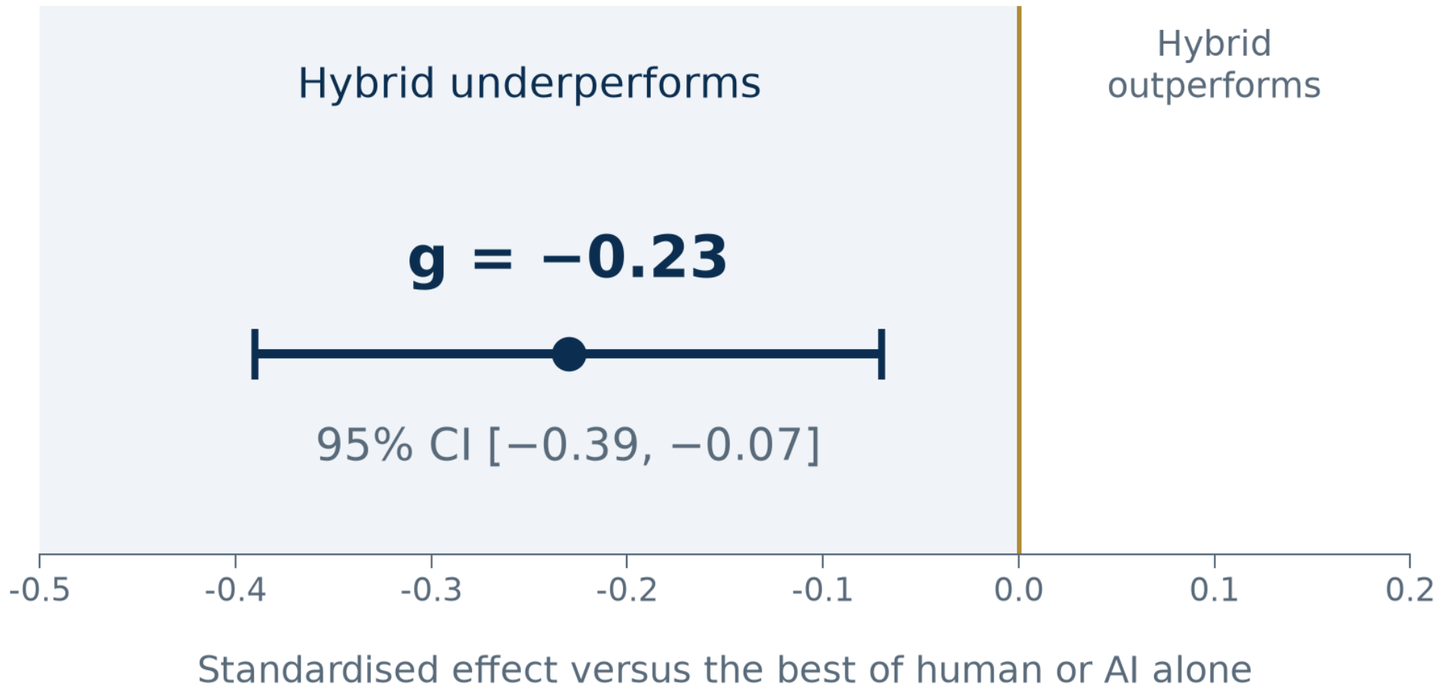
1 / 8 retained legacy Power Peg; full source-code reconstruction is absent.

Source: SEC Order (2013), p. 6 · SEC finding: 7 updated versus 1 non-updated SMARS server, out of 8.

Human + AI: choose the right comparator

106 EXPERIMENTS · 370 EFFECT SIZES · VACCARO ET AL., 2024

Hybrid versus the better standalone option



Compared with humans only

Hybrid performance improved on average. That is augmentation —not proof of synergy.

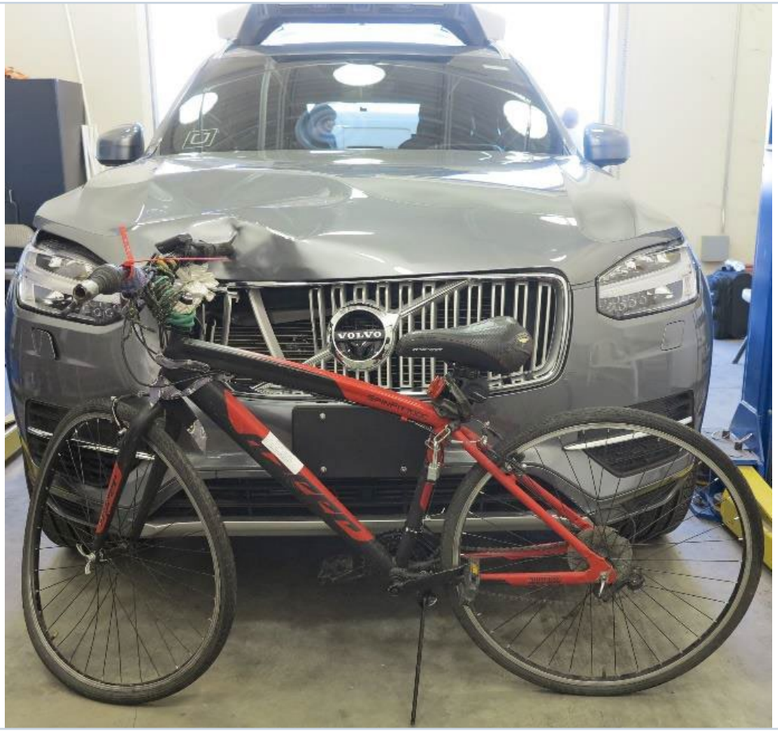
Compared with the best of human or AI

The pooled effect was negative. Results vary across tasks and workflows.

MASAI tests a specific clinical workflow —not a universal hybrid verdict.

Plot redrawn from Vaccaro et al. (2024) · Studies Jan 2020 –Jun 2023 · MASAI (2023) uses a different task and comparator.

Uber Tempe: a sole operator was not a control



NTSB investigation photograph: damaged Volvo XC90 and bicycle after the Tempe collision.

NTSB investigation HWY18MH010

CONTROL TEST: WHAT DID THE SAFETY NET ACTUALLY DO?

01 **One operator was the live fallback.** After an HMI was installed, one operator combined environmental monitoring and event tagging in an ADS-controlled vehicle.

02 **Monitoring data did not ensure oversight.** ATG could retrospectively monitor behaviour and procedure adherence, but rarely used those means.

03 **A vehicle safety layer was removed.** Volvo factory FCW/AEB was disabled because of potential radar interference; NTSB found no full replacement redundancy.

Engineering inference: a person in the loop is a control only when monitoring is feasible, intervention is usable, redundancy remains, and oversight is exercised.

Source: NTSB HAR-19/03 · Findings concern the 2018 developmental-ADS test configuration, not a general ADS standard.

Uber Tempe: label findings, not “bias”

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KEEP OBSERVATION, FORMAL DETERMINATION, AND THEORY LABEL SEPARATE

RECORD

Observed record

The ADS alternated classifications among vehicle, bicycle, and other.
Reclassification did not retain tracking history.
Video documented extended off-road gaze before impact.

INVESTIGATOR

NTSB determination

Probable cause: the operator failed to monitor the road and ADS while visually distracted by a personal cell phone.
ATG safety-risk assessment, oversight, complacency mechanisms, and safety culture contributed.

INTERPRETATION

Bound of the claim

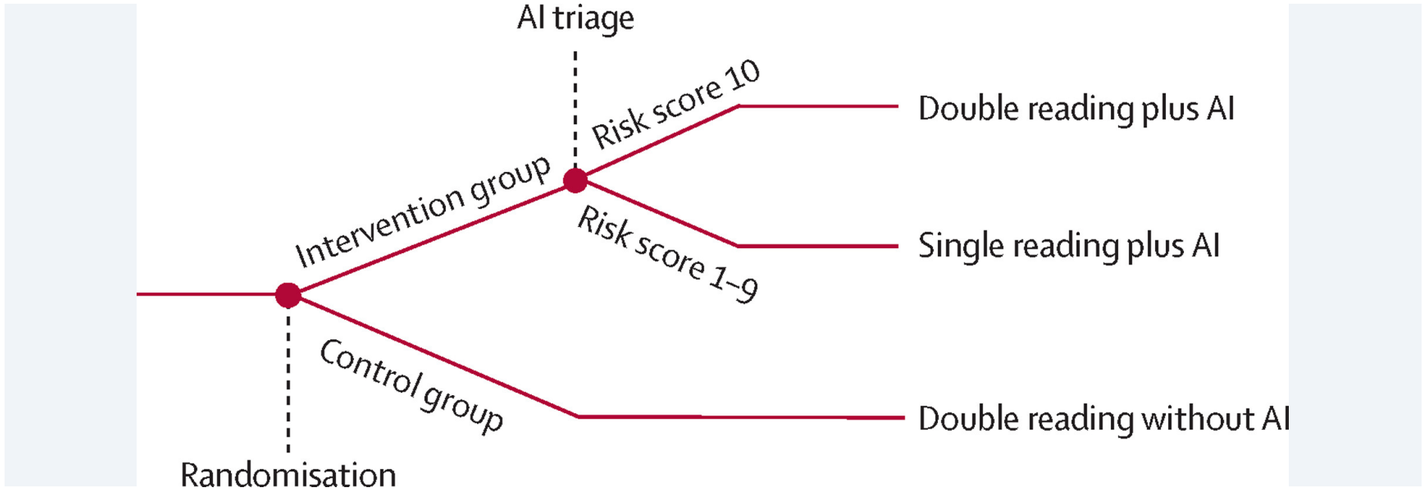
“Automation complacency” is part of NTSB's organizational finding.
The report does **not** make a clinical or cognitive-bias diagnosis of the operator.

Use theory to create testable design questions—such as how to prevent passive monitoring—not to rename formal findings with an unsupported diagnosis.

Source: NTSB investigation HWY18MH010 · Formal probable cause and contributing factors; no clinical bias diagnosis.

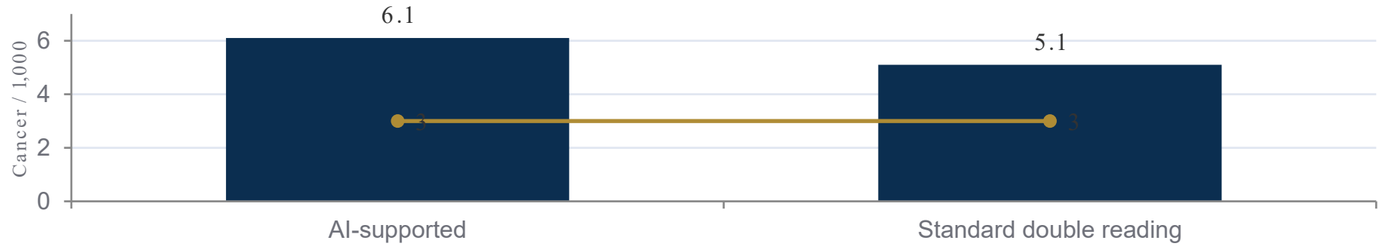
MASAI: a live randomised clinical -safety analysis

ORIGINAL WORKFLOW FIGURE · LÅNG ET AL. (2023)



Reproduced study workflow: randomisation → AI triage → single or double reading; control → double reading without AI.

Interim cancer -detection rate (per 1,000 screened)



Trial frame

80,033 women, 40–80 years, at four Swedish screening sites. PACS automatically allocated completed examinations 1:1 after image acquisition.

Masking and analysis

Participants and radiographers were masked; readers were unmasked. Modified ITT: 39,996 vs 40,024; 13 exclusions.

Interpret the interim result narrowly.

Rate ratio 1.2 (95% CI 1.0–1.5; p=0.052): authors described detection as similar, not proven superiority.

Gate: >3/1,000; observed 6.1/1,000. Primary interval-cancer follow-up pending; not a mortality result.

Source: Lång et al. (2023), MASAI clinical-safety analysis · 2023 interim evidence only.

The paradox: harder work for the human

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BAINBRIDGE'S IRONY OF AUTOMATION · AIR FRANCE 447



F-GZCP (AF447 aircraft), CDG, 2007 · P. Kierzkowski / CC BY -SA 3.0

Automation handles routine work. Humans diagnose and recover.

1 · Evidence becomes unreliable
Temporary inconsistent airspeed measurements.

2 · Automation disengages
Autopilot off; control and diagnosis return to the crew.

3 · Coordinated recovery
Stall not identified; training, interfaces and coordination mattered.

Optional: BEA flight animation →

Multi-factor accident —not evidence that AI caused a crash.

Sources: Bainbridge (1983), pp. 775–777 · BEA Final Report (2012), §3.2 · Photo credit above.

AF447: analyse a handover as a multi-factor chain

BEA FINAL REPORT (2012) · SPEED-MEASUREMENT SYSTEM AND HANDOVER ANALYSIS

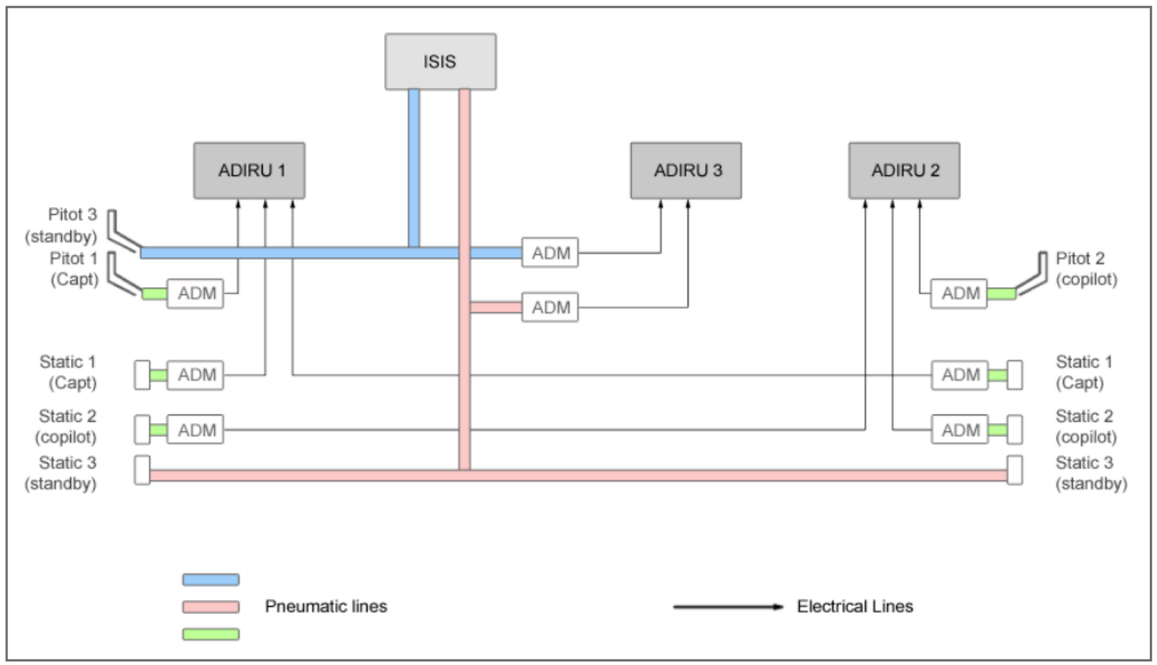


Figure 4: Diagram of the speed measurement system architecture

BEA Figure 4 · A330 speed-measurement system architecture (report p. 33).

The technical signal path is only one part of a safe handover.

1 · Signal integrity
Independent channels can still yield an inconsistency to interpret.

2 · Mode transition
Autopilot disconnection shifts control and diagnosis to the crew.

3 · Recovery boundary
Recognition, pitch/thrust control and coordination must occur together.

Analyse across **training · interface/modes crew coordination · operational context**

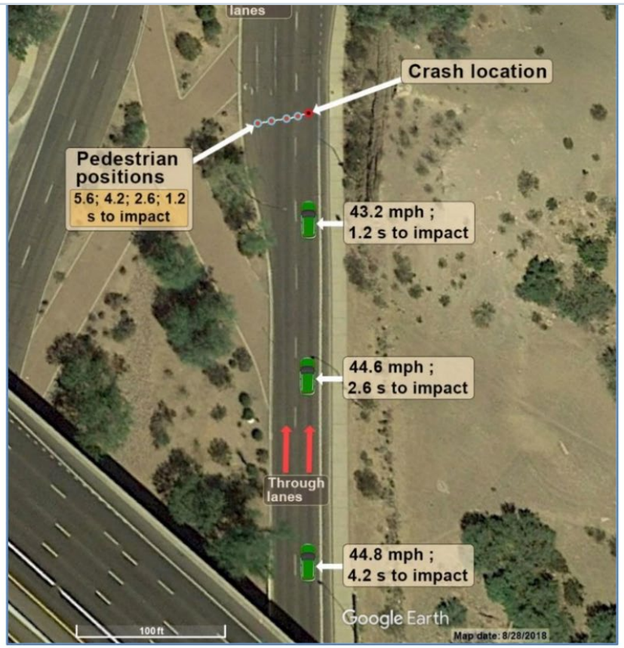
Design question: are mode-change cues salient, interpretable and rehearsed for safe recovery?

Source: BEA Final Report (2012), Figure 4, p. 33; §3.2.

Uber Tempe: braking -handover timeline

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NTSB REPORT FIGURE · CRASH PATH BEFORE IMPACT



NTSB HAR-19/03 Figure 1; adapted from Google Earth.

DOCUMENTED CRASH SEQUENCE · SECONDS BEFORE IMPACT

−5.6 s — **Object detected.** The ADS never accurately classified the person as a pedestrian or predicted her path.

−1.2 s — **Collision imminent.** At −1.2 seconds, the ADS judged a crash imminent after the earlier classification and prediction failures.

1.0 s — **Action suppression:** braking was withheld for reassessment or takeover, with no operator alert when suppression began.

−0.02 s — **Late steering input.** ADS data recorded left steering at 39 mph, 0.02 seconds before impact.

Impact · 0 s

Redundancy context: Volvo FCW/AEB was disabled during testing because of potential radar interference. NTSB found no full replacement safety layer; Volvo simulation said it **might** prevent or mitigate a crash.

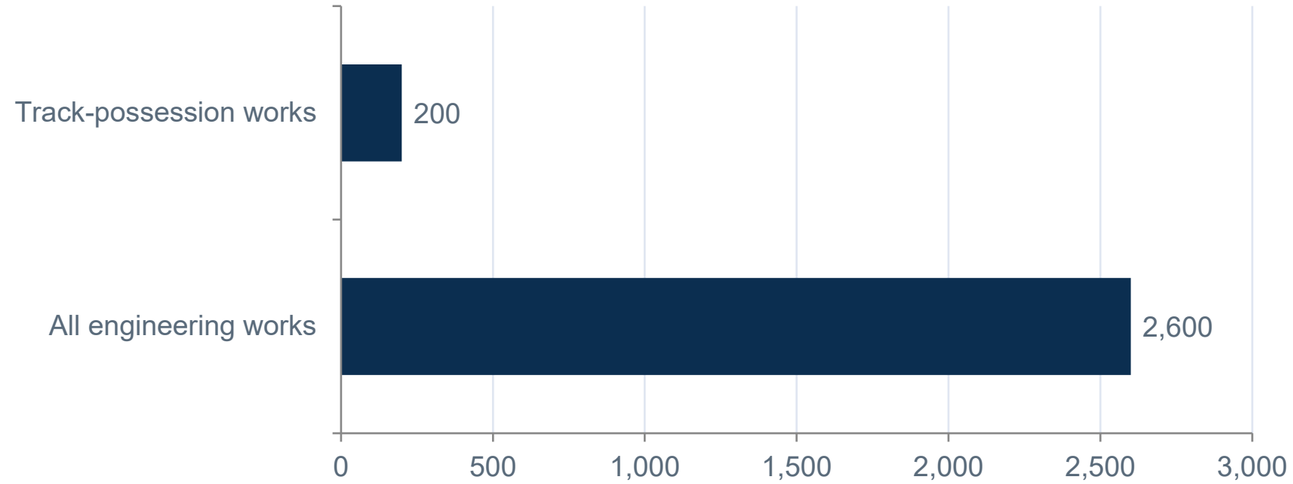
Source: NTSB HAR-19/03 · Recorded sequence; it does not show that any one alternative control would certainly prevent this crash.

MTR ETMS: workload and constraint handling

REPORTED REDESIGNED DEPLOYMENT SCALE · CONDITIONS, NOT A PERFORMANCE BENCHMARK

Minimum reported weekly counts

The bars display reported lower bounds; track-possession works are a subset, not a comparable outcome measure.



More than 10,000 maintenance personnel were involved at the reported deployment scale.

Roughly 4 –5 hours
nightly non-traffic window for engineering works

Documented constraint categories

Possession segments
Non-overlap; power compatibility.

Resources
Wagons, locos and operators.

Network operations
Train paths and noise limits.

Rules and records
Safety, operational, statutory.

Reported: >30 minutes saved per nightly dispatch.
No baseline duration, observation period or independent audit is given.

Source: MTR ETMS implementation account (IAAI/AAAI, 2014) · Scale and dispatch-time claims are reported, not independently audited.

Uber Tempe: questions for accountable control

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4 MINUTES · ANALYSE THE PUBLISHED NTSB INVESTIGATION

TWO EVIDENCE-LED QUESTIONS

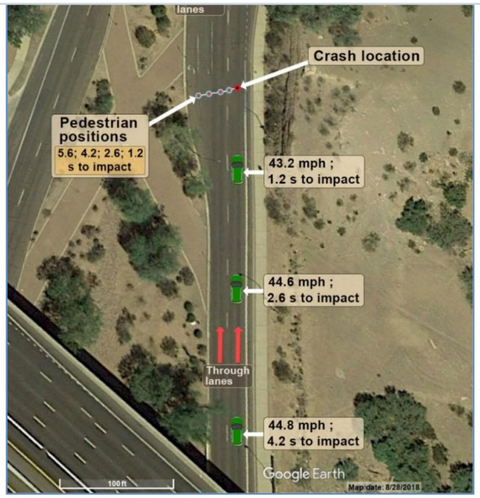
01 Consolidating supervision

What safety-assurance evidence should have been required before one operator assumed both environmental-monitoring and event-tagging work?

02 Suppressing emergency braking

How should responsibility be evidenced when braking is suppressed without an alert and a sole operator is expected to intervene?

NTSB FIGURE 1 · RECORD BEFORE INTERPRETATION



Crash path, pedestrian positions and vehicle movement before impact.

CLASSIFY BEFORE RECOMMENDING

Documented crash-time fact → NTSB determination → Prospective recommendation

Source: NTSB HWY18MH010 · H-19-052 recommended a complete ADS-testing safety-management system; it was not crash-time evidence.

Will machines replace jobs —or reshape them?



CGP Grey · What Happened to Horses Is Happening to Us

Watch on YouTube →

BEFORE VIEWING

If a machine can perform a task, must a person's job disappear?

AFTER VIEWING

Name one assumption behind the substitution argument. What could change its conclusion?

Capability ≠ adoption ≠ job loss
Demand, costs and institutional choices intervene.

Historical provocation (2014), originally "Humans Need Not Apply" · Suggested viewing budget: 3 minutes · Not a current forecast.

Exposure is not observed unemployment

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ILO · GLOBAL OCCUPATIONAL EXPOSURE TO GENERATIVE AI · 2025



Some occupational GenAI exposure

3.3%

Of global employment is in the highest exposure category —a subset of exposed workers.

Task exposure ≠ adoption ≠ job loss

The ILO identifies job transformation as more likely than wholesale replacement.

Clerical occupations remain most exposed.

Source: Gmyrek et al., ILO Working Paper 140 (20 May 2025) · Ring depicts the reported approximate 1-in-4 share; not actual dismissal rates.

Copilot: task evidence is not job evidence

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2022 CONTROLLED EXPERIMENT · COMPLETION ASSISTANT, NOT THE CURRENT CLOUD AGENT

What the experiment actually observes

95 professional programmers recruited through Upwork
45 Copilot treatment · 50 no-Copilot control

One standardised assignment
Implement a JavaScript HTTP server from skeleton code; both arms could use ordinary web resources.

Completion outcome
Repository creation to the first commit passing all tests. **12** provided

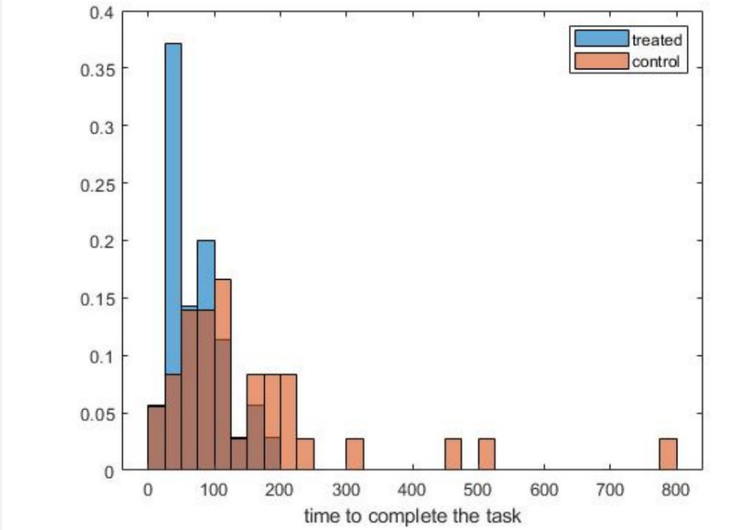


Figure 6: Time to task completion

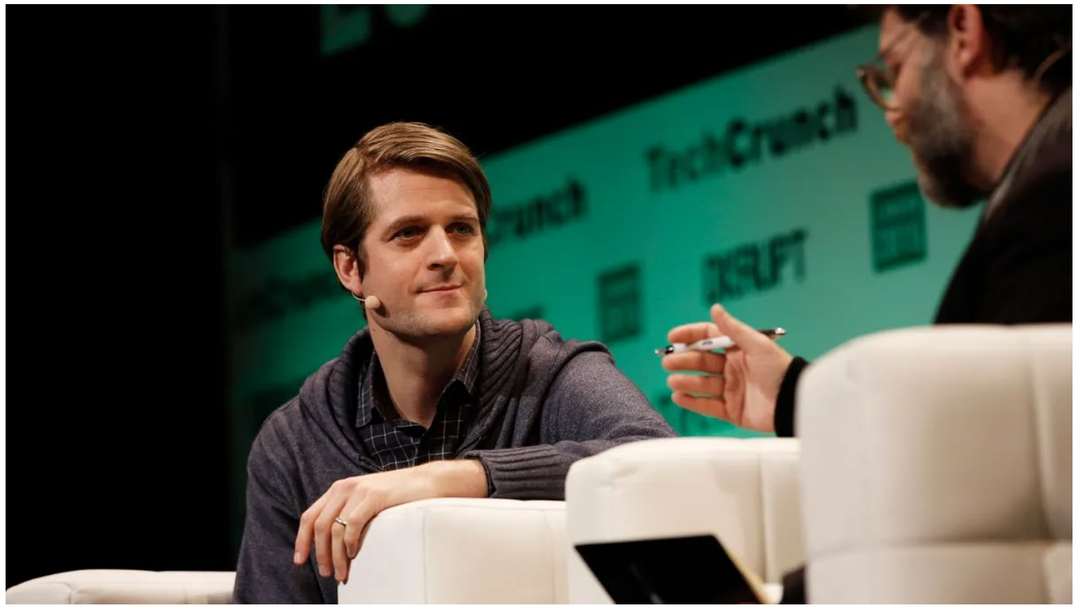
lote: Distribution of time to task completion between treated (blue) and control (orange) groups

Paper Figure 6: task-completion-time distribution (treated / control).

It does not identify headcount, wages, job quality, or economy-wide displacement.

Sources: Peng et al. (2023) · Original Figure 6 reproduced from the paper; one bounded 2022 assignment.

Klarna’s service shift is not a labour forecast



Siemiatkowski (left), TechCrunch Disrupt London 2015—leadership context, not a 2025 service event.
Photo: John Phillips / Getty Images via CX Dive.

TWO EVIDENCE RECORDS—NOT A TRENDLINE

27 February 2024 · company announcement
First-month coverage, capacity, speed, and quality claims; a live-agent choice was stated.
Claims were not independently audited.

8–9 May 2025 · reported service stance
Customers should always have a real-person option; recruitment and a pilot were reported.
This is not an outcome or staffing evaluation.

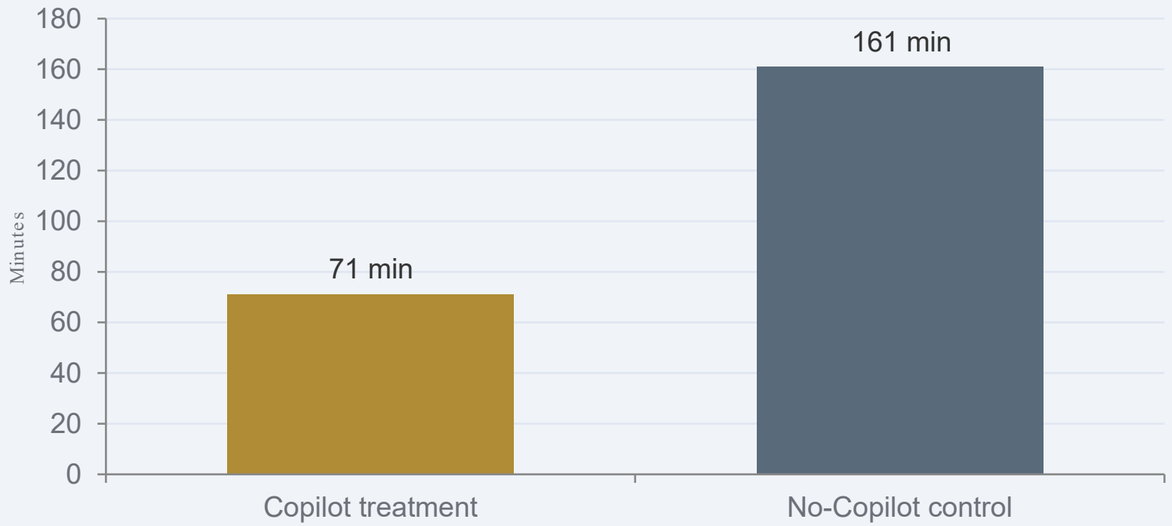
Autor’s task framework explains why neither statement yields a net-employment conclusion.

Substitution · automated tasks	**Complementarity** · changed human tasks	**Output demand** · linked demand changes

Sources: Autor (2015) · CX Dive (9 May 2025) · Later stance is not staffing/outcome evidence.

Copilot: a bounded productivity result

2022 JAVASCRIPT HTTP - SERVER EXPERIMENT · COMPLETION TIME AMONG TASK COMPLETERS



Conditional estimate

55.8% faster

Relative to the control mean, **among task completers** .
95% CI for improvement: **21%–89%** .
Four >300-minute outliers were control; the result remained robust when omitted.

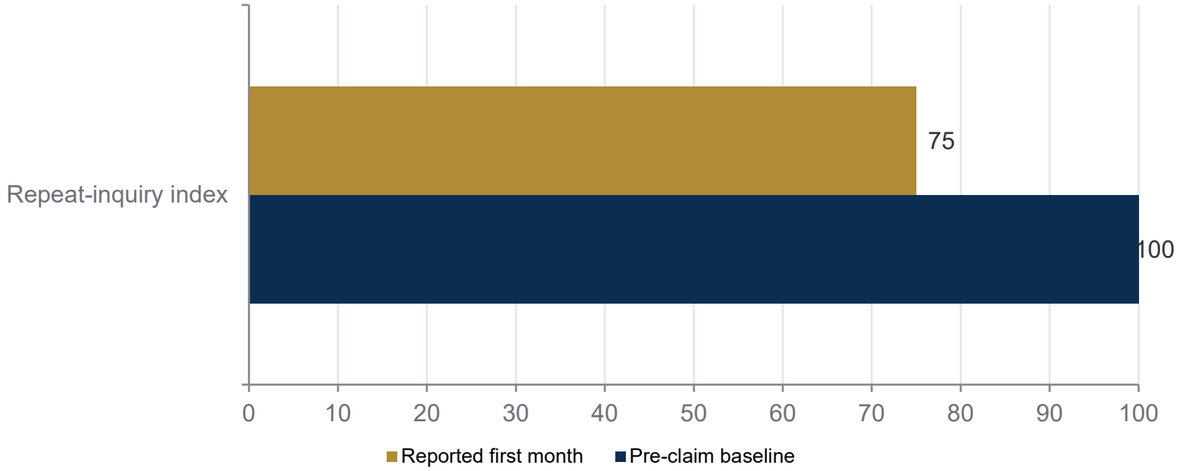
Do not infer: code quality, security, team integration, maintenance, or current-agent productivity.

Source: Peng et al. (2023), Figure 6 and results · Task success was +7 points; 95% CI −11 to +25, not statistically significant.

Klarna’s reported gains leave distribution unmeasured

First-month figures are company-reported; no independent audit or underlying dataset is supplied.

Repeat-inquiry claim, normalised for comparison



Index sets the undisclosed baseline to 100; the 75 index expresses Klarna’s reported 25% reduction.

What Klarna reported

Coverage and capacity: 2.3 million conversations; two-thirds of chats; work equal to 700 FTEs.
Speed and quality: less than 2 minutes versus 11; satisfaction “on par”.

What remains undisclosed

Counting rules, FTE formula, task mix, samples, denominator, time window, score method, and causal comparison.

Operational metric ≠ distributional outcome. The record does not establish worker allocation, pay, conditions, welfare, or who received any gain.
Compare AI-only, human-assisted, and human-resolved outcomes before claiming shared gains.

Source: Klarna (27 Feb 2024) press announcement · Qualification reflects undisclosed methods and workforce data.

A human option is not a documented hand-off design

The public record states a human-service option; it does not specify the customer journey into that option.

DOCUMENTED IN THE 2024 ANNOUNCEMENT

Klarna app assistant
Refunds, returns, payment issues; cancellations, disputes, and invoice inaccuracies.

→

Live-agent choice
Customers could choose a live agent, according to the February 2024 release.

NOT PUBLICLY DESCRIBED AS A HAND-OFF MECHANISM

Routing criteria · market and language coverage · availability · wait time
Actual hand-off path · uptake · resolution or quality outcomes

Reported CEO stance · May 2025
Customers should always be able to speak to a real person; recruitment and a pilot were reported.

Evidence boundary
A nominal human presence is not evidence of usable recourse or verified performance.

Recommended engineering lesson —not a documented Klarna control: specify and test a low-friction human-access route. Report availability and outcomes, especially for unresolved or high-stakes contacts.

Sources: Klarna (2024) · CX Dive (9 May 2025) · 2025 stance is not audited implementation evidence.

Copilot study: speed, not deskilling evidence

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2022 EXPERIMENT · DO NOT TURN A SHORT-TASK COMPLETION RESULT INTO A LEARNING CLAIM

Observed measurement path

One JavaScript HTTP -server assignment

↓

First commit passing **all 12** provided tests

↓

Observed outcomes
Task success and elapsed completion time

Outside the study protocol

Code quality and security

Long-term learning, retention, or deskilling

Collaboration, review, and maintenance

Career and job outcomes
outside a short, individual task

Engineering lesson: retain review and testing before merge; that is code assurance, not evidence that skills are preserved.

Sources: Peng et al. (2023) · GitHub Docs: review and test before production.

MTR ETMS: Hong Kong rail engineering work

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DOCUMENTED DEPLOYMENT MILESTONES · THE 2014 ACCOUNT DOES NOT VERIFY CURRENT CONFIGURATION



Engineering work with an engineering train — 2014 paper Fig. 2; context image, not a current ETMS screen.

Recorded path to the 2013 redesign

- **2002** · ETMS implemented.
- **Jul 2004** · Original AI Engine deployed.
- **Nov 2012** · ETMS operated without AI while the knowledge base underwent UAT on actual operational data.
- **May 2013** · AI ran in parallel.
- **Jul 2013** · Redesigned engine fully deployed.

Evidence boundary

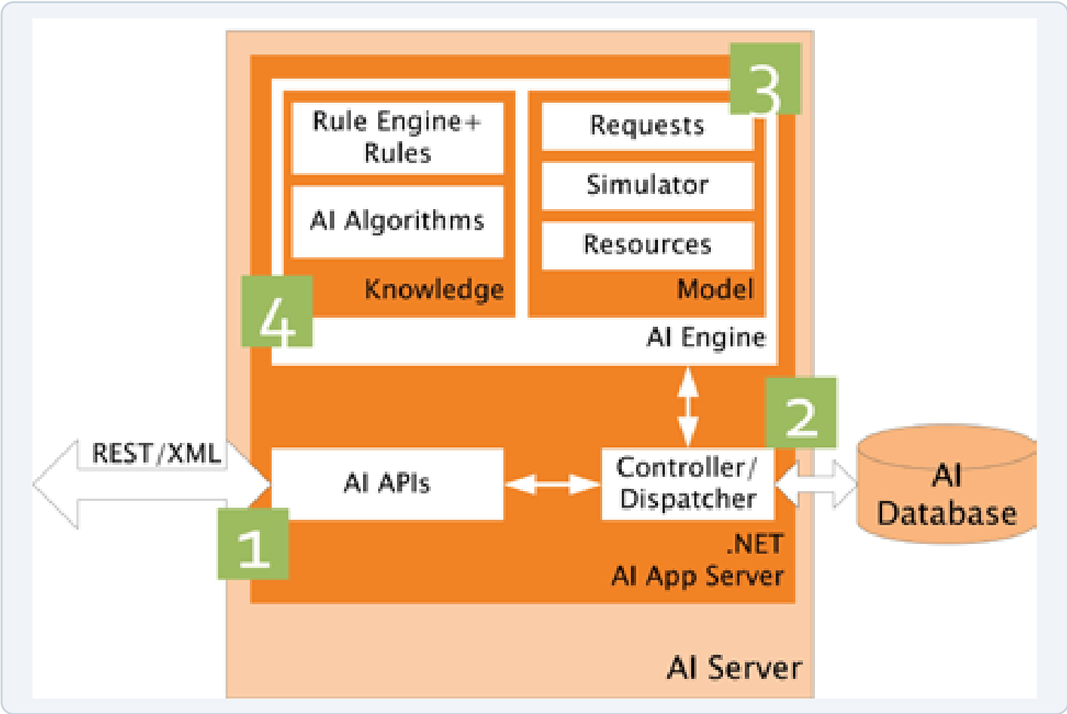
UAT, refinement and a parallel run are documented. No public quantitative pass criteria or error rate.

Source: MTR ETMS implementation account (IAAI/AAAI, 2014) · Milestones are retrospective deployment reporting.

MTR ETMS: read the implementation evidence

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HOW TO DISTINGUISH DOCUMENTED IMPLEMENTATION FROM UNREPORTED ASSURANCE



Detailed AI-server architecture, 2014 paper Fig. 4 — a published design figure, not an assurance case.

Read a system figure in three evidence layers

1 · Published mechanism
Rule engine, rules, requests, simulator, resources and APIs are part of the reported design.

2 · Documented process
UAT used actual operational data; gaps and parameters were refined; a May 2013 parallel run preceded deployment.

3 · Evidence gap — do not reverse it
Full rule mapping, UAT coverage/error rate, approver identity and reschedule re-approval are not reported.

Evidence-led question · What traceability, coverage and near-miss evidence would make encoded rules examinable?

Source: MTR ETMS implementation account (IAAI/AAAI, 2014) · Unreported details are evidence gaps, not alleged missing controls.

MTR ETMS: what the record supports

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DOCUMENTED PRACTICE ≠ PROVEN EFFECTIVENESS ≠ COMPLETE GOVERNANCE RECORD

Documented implementation
Human planners vet the AI-generated weekly schedule for final approval before execution.

Careful conclusion
UAT on actual operational data and a parallel run show testing/refinement before full July 2013 deployment.

Limit to preserve
No public UAT coverage, pass threshold, error rate, approver identity, escalation path or sign-off for every reschedule.



MTR "Super OCC", 2014 paper Fig. 5 — control-centre context, not proof of current configuration.

Context photo: evidence comes from the paper

Prospective engineering lesson
Record rule changes, validation coverage and exception outcomes.

Prospective engineering lesson
Record authority and rationale when a late change moves from feasible to authorised.

Source: MTR ETMS implementation account (IAAI/AAAI, 2014) · Bottom-row lessons are recommendations, not MTR claims.

Robodebt: an average became an allegation

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AUSTRALIA · ROYAL COMMISSION REPORT, 2023



Report volumes. Image: Guardian (2023)

Sources: Royal Commission report (2023) · Image: Guardian (2023)

PRIMARY INSTITUTIONAL RECORD

The report reconstructs an evidentiary failure.

An employer-reported annual income total entered the data-matching process.
The scheme then treated an average across fortnights as evidence for a debt allegation.

Annual income does not establish income in each fortnight.
The defect is evidentiary.

Rule-based institutional automation; not generative AI.

Robodebt: valid data, invalid inference

The scheme compared quantities that did not answer the same eligibility question.

OBSERVED QUANTITY

Annual income total

A correct annual total can still conceal when income was actually earned.

→

AUTOMATIC PROXY

Annual total averaged across assessment periods



One period-average proxy replaces a time-specific record.

Conceptual mechanism; no constructed recipient data.

→

RELEVANT EVIDENCE

Income actually earned in each assessment fortnight

Entitlement depended on period-specific income; averaging could not supply that record.

Correct data does not guarantee correct evidence.
Validate the target and evidence model—not only the data pipeline.

Source: Royal Commission report (2023) · Mechanism is a documented process schematic.

Robodebt: automation moved the proof burden

Recipients were asked to reconstruct income history to contest debt allegations based on averaging.



Catherine Holmes at the Robodebt hearing — ABC News

The inference survived; the proof moved.

An accessible challenge route must be usable —not merely present.

Institutional record	Averaged allegation	Proof shifted
Annual record held. Data matching flags a discrepancy.	Average cannot establish income by fortnight.	Past records are needed to contest the inference.

Administrative efficiency can become work imposed on the affected person.
A challenge route must be usable, not merely available.

Sources: Royal Commission report (2023), Recommendations 17–18 · Image: ABC News

Robodebt: what the Commission recommended

Read the recommended controls as responses to an investigated institutional failure.

A recommendation is not an implemented control.

The proposed response spans the decision, its rules, institutional monitoring and recovery practice.

TEACHING DISTINCTION

A report recommendation identifies what should be built; it does not prove that it was operating.

RECOMMENDED CONTROL STACK

REC. 17.1	Explain automated decisions; provide a clear review path.
REC. 17.1	Open business rules and algorithms to expert scrutiny.
REC. 17.2	Create oversight of automated government decision-making.
REC. 18.1	Consider dignity, proportionality and review before recovery.

Recommendations do not establish implementation.

Source: Royal Commission report (2023), Recommendations 17.1, 17.2 and 18.1 · Paraphrased.

Hong Kong: governance around the whole system

PCPD's model framework is guidance—not a new AI Act. Applicable PDPO duties remain relevant.

1 · Strategy and governance

Name responsibilities. Document the intended use before deployment.

2 · Risk assessment and oversight

Match the level of human involvement to risk.



3 · Model customisation and management

Manage system behaviour across implementation and operation.

4 · Stakeholder engagement

Communicate how the system is used. Engage and revise.

Higher risk
Human-in-the-loop

Minimal / low risk
Human-out-of-the-loop

If neither fits
Human-in-command

Adapted visual summary: PCPD Model Framework (2024), §2.2–2.3 and pp.44 –45 · Human presence alone does not establish effective oversight.

When an assistant acts as you



Google Duplex · Google I/O 2018 appointment - booking demo

Watch the YouTube demonstration →

Repost by Jeff Grubb's Game Mess. Suggested viewing budget: 3 minutes.

User Sets the goal → **Assistant** Calls → **Business** Responds

Whose consent —and whose authority —does the agent carry?

BENEFIT

Convenience and access for people facing language or hearing barriers.

BOUNDARY

Disclose the agent's identity. Limit commitments. Provide a human fallback.

A successful booking does not settle the ethics of how the interaction occurred.

Source: Leviathan & Matias, Google Research (2018) · Narrow-domain demo; not a general agent. Google discussed transparency.

Copilot coding agent: permission defines authority

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CURRENT GITHUB DOCUMENTATION · DISTINCT FROM THE 2022 COMPLETION-ASSISTANT EXPERIMENT



Official GitHub product visual: assign a GitHub task to Copilot (2025).

Authority is a release path, not an output guarantee

Start: Agents panel, Chat, issue, or PR delegates a task.

↓

Work: research, plan, read/edit code, and run tests/linters in an ephemeral cloud environment.

↓

Scope: one repository · one branch · one PR; no direct push to default.

↓

Actions gate: agent-triggered workflows need write access - user approval.

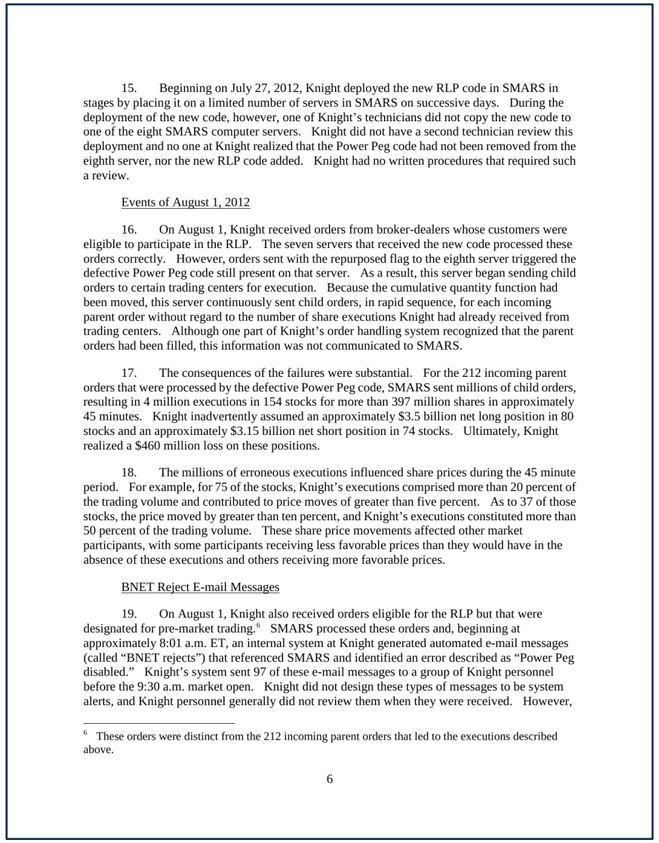
Recommendation, not proof of safety: human review and test before merge.

Sources: GitHub Docs: cloud-agent workflow and limits · GitHub Blog product visual (2025).

Knight: safeguards separated by evidence status

DO NOT TURN A FINDING OF ABSENCE INTO A CLAIM OF IMPLEMENTATION

SEC ORDER · ORIGINAL PAGE 6



15. Beginning on July 27, 2012, Knight deployed the new RLP code in SMARS in stages by placing it on a limited number of servers in SMARS on successive days. During the deployment of the new code, however, one of Knight's technicians did not copy the new code to one of the eight SMARS computer servers. Knight did not have a second technician review this deployment and no one at Knight realized that the Power Peg code had not been removed from the eighth server, nor the new RLP code added. Knight had no written procedures that required such a review.

Events of August 1, 2012

16. On August 1, Knight received orders from broker-dealers whose customers were eligible to participate in the RLP. The seven servers that received the new code processed these orders correctly. However, orders sent with the repurposed flag to the eighth server triggered the defective Power Peg code still present on that server. As a result, this server began sending child orders to certain trading centers for execution. Because the cumulative quantity function had been moved, this server continuously sent child orders, in rapid sequence, for each incoming parent order without regard to the number of share executions Knight had already received from trading centers. Although one part of Knight's order handling system recognized that the parent orders had been filled, this information was not communicated to SMARS.

17. The consequences of the failures were substantial. For the 212 incoming parent orders that were processed by the defective Power Peg code, SMARS sent millions of child orders, resulting in 4 million executions in 154 stocks for more than 397 million shares in approximately 45 minutes. Knight inadvertently assumed an approximately $3.5 billion net long position in 80 stocks and an approximately $3.15 billion net short position in 74 stocks. Ultimately, Knight realized a $460 million loss on these positions.

18. The millions of erroneous executions influenced share prices during the 45 minute period. For example, for 75 of the stocks, Knight's executions comprised more than 20 percent of the trading volume and contributed to price moves of greater than five percent. As to 37 of those stocks, the price moved by greater than ten percent, and Knight's executions constituted more than 50 percent of the trading volume. These share price movements affected other market participants, with some participants receiving less favorable prices than they would have in the absence of these executions and others receiving more favorable prices.

BNET Reject E-mail Messages

19. On August 1, Knight also received orders eligible for the RLP but that were designated for pre-market trading.[6] SMARS processed these orders and, beginning at approximately 8:01 a.m. ET, an internal system at Knight generated automated e-mail messages (called "BNET rejects") that referenced SMARS and identified an error described as "Power Peg disabled." Knight's system sent 97 of these e-mail messages to a group of Knight personnel before the 9:30 a.m. market open. Knight did not design these types of messages to be system alerts, and Knight personnel generally did not review them when they were received. However,

[6] These orders were distinct from the 212 incoming parent orders that led to the executions described above.

6

INTERPRET THE RECORD BY STATUS

Documented but inadequate	Erroneous-order controls existed upstream. SEC found the final market-entry point at SMARS insufficiently controlled.
Actual missing / deficient control	SEC found no adequate SMARS output monitoring or halt procedure, and no aggregate firm-wide thresholds linked to automatic order prevention.
Engineering lesson	Prospectively: independently verify every deployment target and turn deployment-failure signals into owned alerts with a response path.

The third row is a lesson derived from findings, not a claim about Knight's 2012 practice.

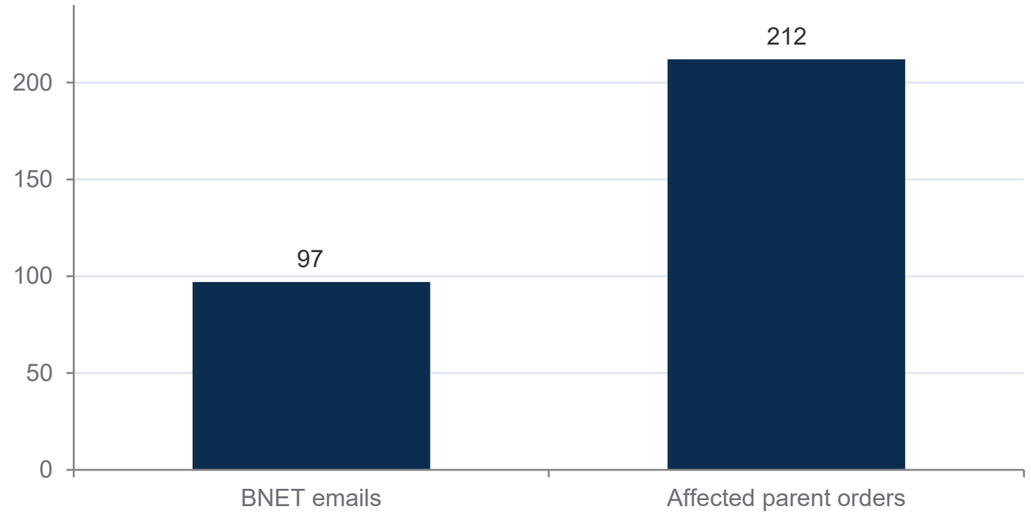
Page 6: deployment and BNET rejects.

Source: SEC Order (2013), pp. 6 –8, 12 · Settled findings and potential preventative measures, not an implementation record.

Knight: an email is not an operational alert

PRE-OPEN SIGNALS EXISTED · THE RESPONSE SYSTEM DID NOT

DISTINCT COUNTS IN THE SEC RECORD



Different units: 97 emails ≠ recipients or orders; 212 is the affected-parent-order denominator.

About 45 minutes after open: 4 million executions in 154 stocks.

FROM NOTIFICATION TO SAFETY CONTROL

c. 8:01 ET
97 rejects

Before open
Not an alert

Post-open
Orders flow

The emails were generally not reviewed on receipt, not acted on before open, and not used later to diagnose the incident.

ENGINEERING LESSON · REQUIRED EVIDENCE TRAIL

Signal captured → **Owner** assigned → **Urgency** classified → **Stop** decision

This response path is prospective engineering guidance, not a description of Knight's 2012 logging practice.

Source: SEC Order (2013), pp. 6–7 · 97 is the reported email count, not recipients or affected orders.

Knight: release gates derived from SEC findings

The SEC record yields **testable release gates** —not proof that these controls were in place.

RELEASE GATES FOR ANY MARKET-FACING CHANGE

1 · Server completeness

Record: 7 of 8 servers were updated; no required second-technician review.

Can an independent check attest that every production server matches the release manifest?

IF NOT: HOLD ACTIVATION

2 · Legacy-path retest

Record: Power Peg was discontinued in 2003; a 2005 change was not retested if called.

Could a retired callable path or activation sequence still be reached?

IF YES: RETEST BEFORE RELEASE

3 · Output and stop

Record: SEC found SMARS output monitoring, thresholds, and halt procedures inadequate.

Can market-facing output be monitored, prevented at thresholds, and disconnected?

IF NOT: DO NOT ACTIVATE

Release decision — engineering lesson: a successful feature test is not sufficient evidence that delegation is safe.

Source: SEC Order (2013), pp. 5–8 · Gates are derived lessons, not SEC-prescribed wording or historical Knight controls.

Three ethical lenses applied to actual cases

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A factual finding and an ethical judgement answer different questions.

Consequences · AF447

The investigation identified a chain involving sensors, flight - control transition, crew actions and stall recognition.

↓

Assess the whole safety chain, not routine reliability alone.

Rights and duties · Duplex

A system represented a user while interacting with a business. Google discussed transparency and fallback.

↓

Successful execution does not settle disclosure and authority.

Care and justice · Robodebt

Income averaging placed evidentiary work on recipients. The Commission called for review and humane recovery.

↓

Efficiency must account for the burden placed on people.

Engineering lessons are reasoned conclusions —not claims that every safeguard was implemented.

Case sources: BEA (2012) · Google Research (2018) · Robodebt Commission (2023)

Evidence audit: Klarna’s published claim package

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Read the announcement as a company-reported claim package—not an independent employment or quality study.

ATTRIBUTION LEDGER

Published claim	What the cited release establishes	What an evidence audit must not infer
2.3 million conversations; two-thirds of chats	Klarna’s first-month global deployment claim	Audited chat counts, counting rules, or general service performance
Work equal to 700 full-time agents	A workload-equivalence estimate against human-agent capacity	700 jobs removed; staffing, pay, or working conditions
Satisfaction “on par”; 25% fewer repeats; less than 2 versus 11 minutes	Company-reported service and efficiency claims	Score method, sample, denominator, task mix, causal effect, or external validation

Evidence-led question: which claim can be attributed to Klarna, and what missing measurement would be needed before a workforce or quality conclusion?
The reported 2025 human-service stance does not fill these 2024 measurement gaps.

Sources: Klarna (27 Feb 2024) · CX Dive (9 May 2025).

Uber Tempe: causal responsibilities in the NTSB record

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6-MINUTE EVIDENCE ANALYSIS · ATTRIBUTE FINDINGS WITHOUT INVENTING STAKEHOLDER VIEWS

ASSIGN EACH FORMAL FINDING TO ITS CONTROL BOUNDARY

Vehicle operator · monitoring

NTSB probable cause: failure to monitor the driving environment and ADS while visually distracted by a personal cell phone.
Keep this distinct from a clinical diagnosis.

Uber ATG · safety assurance

Contributing conditions: inadequate safety-risk assessment, ineffective operator oversight, inadequate complacency mechanisms, and safety culture.

ADS and vehicle · design

Reclassification discarded tracking history; braking was suppressed without an alert; FCW/AEB was disabled without full replacement redundancy.

ANALYSIS SEQUENCE

Report finding → monitoring, safety assurance, or technical design → evidence needed to assess that boundary

Source: NTSB investigation HWY18MH010 · Uses report findings, not fictional stakeholder statements; no one control is claimed to certainly prevent the crash.

Assess MASAI evidence: four questions before transfer

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Use the 2023 interim analysis to appraise evidence —not to issue a global safety certificate.

1 · Is the comparison credible?

Record: 80,033 women; automatic 1:1 allocation after image acquisition; contemporaneous double-reading comparator.

Keep visible: reading radiologists were unmasked.

2 · What was delegated?

Record: score 1–9 routed to single reading; score 10 to double reading; radiologists decided recall.

Keep visible: this threshold is not a universal deployment rule.

3 · Which outcomes were observed?

Record: 44.3% fewer readings; 6.1 versus 5.1 cancers / 1,000; false positives 1.5% in each arm.

Keep visible: interim analysis; no mortality result.

4 · Can this protocol transfer?

Record: four Swedish sites; participants aged 40–80.

Keep visible: no race/ethnicity data; other radiologists, AI versions, equipment, and populations need prospective study.

Evidence-led question: what result —or missing outcome —would change the triage boundary?
Passing the >3/1,000 interim safety floor supported continuation, not universal safety or superiority.

Sources: Lång et al. (2023) · ClinicalTrials.gov NCT04838756.

Automation does not erase answerability

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Delegation changes the work.
It does not remove the duty to justify it.

Source
What is documented?

→

Workflow
Who did what?

→

Outcome
What was measured?

→

Judgement
What is justified?

Exit analysis: choose one real case from the lecture.
Identify a documented control or failure, an evidence limit, and the ethical conclusion that follows.

BEA: AF447 investigation · Robodebt Commission report
PCPD: Hong Kong model framework · Vaccaro et al.: human–AI evidence

Course policy: develop your own analysis and acknowledge permitted AI collaboration; do not submit AI-generated assignment content.